🧱 PinResQ – Technology Stack

🎨 Frontend (Client Side)

- **React.js (JavaScript)**
  - Used to build the user interface
  - Handles pages like Login, Dashboard, Map, etc.
- **HTML5 + CSS3**
  - Structure and styling of UI
  - Responsive design
- **Axios**
  - Handles API communication between frontend and backend
- **React Router DOM**
  - Enables navigation between pages
- **Leaflet.js**
  - Map integration using OpenStreetMap
  - Displays real-time accident locations with markers

⚙️ Backend (Server Side)

- **Java**
  - Core programming language
- **Spring Boot**
  - Backend framework to build REST APIs
  - Handles business logic and request processing
- **Spring Data JPA (Hibernate)**
  - ORM tool for database interaction

    - Converts Java objects ↔ database tables

🗄️ **Database**

- **MySQL**
    - Stores user data and emergency reports
    - Structured relational database

🔗 **API Communication**

- **REST APIs (JSON-based)**
    - Connect frontend and backend
    - Data exchange in JSON format

🌐 **Map & Location Services**

- **Browser Geolocation API**
    - Fetches user's real-time location (latitude, longitude)
- **Leaflet + OpenStreetMap**
    - Displays map and markers
    - Free alternative to Google Maps

🔐 **Authentication (Current + Future)**

- **Basic Authentication**
    - Email + Password login system
- **(Future Upgrade) JWT Authentication**
    - Secure token-based authentication

🧪 Development Tools

- **Node.js + npm**
  - Manage frontend dependencies
- **Maven**
  - Build and run Spring Boot backend
- **MySQL Command Line / Workbench**
  - Database management

🖥️ Development Environment

- Runs on:
  - Command Prompt / Terminal
  - Browser (localhost)

🧠 Architecture Summary

React (Frontend)

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Axios (API Calls)

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Spring Boot (Backend)

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JPA / Hibernate

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MySQL Database